

The code for \$a_2\$ is:

Please provide a brief explanation of your answer.

The number a_2 is defined as the limit of the sequence output by `Compute_a_2( $n$ )`. Inside `Compute_f( $n$ )`, the initial term and the subtracted terms involve $72^{-10^{24}}$, which is unbelievably tiny. The dominant part is the geometric sum $\sum (1/2)^k$. As $n \rightarrow \infty$,

$$f(\infty) = \frac{1}{4} + 72^{-10^{24}} - \sum_{k=3}^{\infty} 72^{-10^{24}k},$$

so

$$a_2 = 2f(\infty) + \frac{1}{2} = 1 + 2 \cdot 72^{-10^{24}} - \frac{2 \cdot 72^{-3 \cdot 10^{24}}}{1 - 72^{-10^{24}}} > 1.$$

The optimization problem is to minimise $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$. Since $a_2 > 1$, the minimum occurs at $x_1 = 0$, $x_2 = 1/a_2$.

Because $a_2 = 1 + \varepsilon$ with ε astronomically small, $1/a_2 \approx 1$ to far more than two decimal digits. Hence an optimal minimiser accurate to at least 2 digits in the ℓ^∞ norm is

$$x_1 = 0, \quad x_2 = 1.$$